

Central Library
Nitte
Duration: 3 Hours

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Note:

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

12. Which function is used to create a histogram in R?
 A) plot() B) hist()
 C) boxplot() D) scatterplot()
13. Which function in ggplot2 is used for creating scatter plots?
 A) geom_line() B) geom_boxplot()
 C) geom_histogram() D) geom_point()
14. Which R package is commonly used for creating interactive data visualizations?
 A) ggplot2 B) lattice
 C) base D) plotly
15. John wants to create a chart that shows trends in sales for four quarters. What type of chart should he create?
 A) Pivot Chart B) Line Chart
 C) Embedded Chart D) Linked Chart
16. Tables created through Excel's table feature allow users to filter _____ by different values.
 A) Rows B) Columns
 C) Sections D) Values
17. Which of the following is Not one of Excel's what-if functions?
 A) Goal seek B) Solver
 C) Scenario manager D) Auto Outline
18. How can we provide a dynamic range in the 'data Source' of pivot tables?
 A) Create a Named range using the Offset function B) Base the pivot table using the Named range created in Step 1
 C) Both A and B D) None
19. Which function you would use to add several numeric values?
 A) SUM B) COUNTA
 C) AVERAGE D) TOTAL
20. Pivot tables that have been created can easily be turned into graphs or charts by simply highlighting the whole pivot table and choosing a graph or chart that you want to create. Is this true?
 A) True B) False, Pivot tables do not contain enough information for graphs or charts
 C) False, only regular tables may be used to create graphs or charts D) False

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is Data Science? Explain the data scientist roles and responsibilities.	5	L*2	1	1
	b) What are the stages in a data science project and explain them in detail.	6	L2	1	1
	c) Explain the applications of Data Science.	5	L2	1	1
2.	a) Describe the steps involved in data mining when viewed as a process of knowledge discovery.	8	L2	2	1
	b) What is a Data Warehouse? explain the key features of OLTP and OLAP system and differentiate between them.	8	L2	2	1
3.	a) What is classification? Explain how the classification problem can be solved using general approach with a neat figure and example.	8	L2	2	2

- b) Predict a class label using Naive Bayesian classification for the below test data,

$X = (\text{age} = \text{senior}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit rating} = \text{fair})$

Class-Labeled Training Tuples from the *AllElectronics* Customer Database

RID	age	income	student	credit_rating	Class: buys_computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

Unit – II

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|--|----|----|---|---|
| 4. a) Define R programming. Explain any five features of R. | 8 | L2 | 2 | 2 |
| b) Explain the arithmetic operators, comparison operators, logical operations with example in R programming. | 4 | L1 | 3 | 1 |
| c) What are R – Matrices? Write R program to create matrices and perform various matrix operations. | 6 | L2 | 3 | 2 |
| 5. a) Write R program to demonstrate Inline Functions, Passing Arguments to Functions. | 6 | L2 | 3 | 2 |
| b) What is a vector in R? Describe different methods to extract specific elements of a vector in R. | 5 | L3 | 3 | 2 |
| c) Explain Data Frames and how to Create a Data Frame in R by importing dataset from CSV, Excel files. | 6 | L2 | 3 | 2 |
| 6. a) Explain the R program to demonstrate the histograms with example. | 5 | L2 | 4 | 2 |
| b) Write R program to demonstrate Box plots and Bar Charts. Discuss suitable examples. | 6 | L3 | 4 | 3 |
| | 10 | L3 | 4 | 3 |

Unit – III

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|---|---|----|---|---|
| 7. a) Describe Data Mining and Business Intelligence. | 8 | L2 | 5 | 2 |
| b) Illustrate logical functions in excel. | 8 | L2 | 5 | 2 |
| 8. a) Explain the VLOOKUP and HLOOKUP functions with example. | 8 | L2 | 5 | 2 |
| b) Discuss the PivotTables with example. | 8 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
